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Project Report

On

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

Submitted By

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M.Sc. IT (Cyber Security) Semester-III

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C E R T I F I C A T E

T O W H O M S O E V E R I T M A Y C O N C E R N

This is to certify that the following students of M.Sc. IT (Cyber Security) Semester-III have completed their project work titled “**Digital Trapdoor: Multi-layer Security using Encryption & Steganography**” satisfactorily fulfill the requirement of M.Sc. IT (Cyber Security) Semester-III, Department of Computer Science, Ganpat University in the academic year 2025-2026.

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I take this opportunity to humbly express my heartfelt gratitude to all those who have supported and guided us in the successful completion of our project titled **“Digital Trapdoor: Multi-layer Security using Encryption & Steganography.”**

We are sincerely thankful to **Ganpat University** for providing us with this valuable opportunity to develop our project. We also extend our gratitude to the **Department of Computer Science, Ganpat University, Kherva**, for offering an excellent environment and facilities that encouraged and supported our development process.

We express our deep sense of gratitude to our guide, **Prof. Bharat Chaudhary**, for his constant guidance, valuable suggestions, and keen interest at every stage of our project. His encouragement and insightful feedback motivated us to work diligently and complete the project in the right direction.

Finally, we would like to thank all those who have directly or indirectly contributed to the successful completion of this project.

With regards,

Kugashiya Milan

Soni Jay

PREFACE

In today's digital era, the security of sensitive information has become a major concern due to the rapid growth of internet-based communication and cyber threats. Protecting data from unauthorized access, interception, and tampering is crucial for individuals, organizations, and governments alike. Traditional single-layer security methods, such as encryption or password protection alone, are often insufficient to prevent sophisticated attacks.

The project "Digital Trapdoor: Multi-layer Security using Encryption & Steganography" aims to address these challenges by combining two powerful security techniques: cryptography and steganography. Encryption converts confidential data into unreadable ciphertext, ensuring that unauthorized users cannot access the content, while steganography conceals the encrypted data within a digital medium (like an image or audio file), making it invisible to unintended observers. By integrating these two layers, the project provides a robust and secure mechanism for data protection during storage and transmission.

I would like to express my sincere gratitude to my faculty guide, colleagues, and institution for their guidance, support, and encouragement throughout the project. I hope that this work will contribute to a deeper understanding of secure communication systems and inspire further research in the field of information security.

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Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

1. Project Profile

Project Title	Digital Trapdoor: Multi-layer Security using Encryption & Steganography.
Objective	Multi-layer Data Protection
OS	Linux, Ubuntu, Windows
Tool	Open puff, split, WeTransfer
Duration	90 Days
Developed by	Kugashiya Milan 24084321001 Soni Jay 240843210029
Group No.	15
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2. Introduction

In the modern digital world, the protection of sensitive data has become one of the most critical challenges due to the increasing risks of hacking, unauthorized access, and cyberattacks. Every day, large amounts of personal, financial, and confidential information are exchanged online, making it essential to adopt strong and reliable data security measures. Our project, titled “Digital Trapdoor: Multi-layer Security using Encryption & Steganography,” aims to develop a secure system that ensures safe communication and data storage by combining two powerful security techniques — Encryption and Steganography.

Encryption is the process of converting readable data into an unreadable form using cryptographic algorithms, ensuring that only authorized users can access the original content. Steganography, on the other hand, involves hiding the encrypted data inside a digital file such as an image, audio, or video, so that the presence of secret information remains undetectable. Together, these two techniques create a multi-layered security system that provides enhanced protection against data theft and cyberattacks.

The main purposes of this project are:

1. To secure data confidentiality through encryption so that information remains protected from unauthorized users.
2. To hide sensitive data using steganography, making it invisible to potential attackers.
3. To ensure multi-layer protection so that even if one layer is compromised, the other still secures the data.
4. To promote secure communication over public or untrusted networks.
5. To create awareness about modern data protection methods combining cryptography and steganography.

This project demonstrates how the integration of encryption and steganography can offer a more robust and reliable security mechanism. It not only helps in protecting information but also contributes to developing better understanding of cybersecurity practices in real-world digital systems.

2.1. Overview

In today's digital era, the protection of sensitive information has become a critical challenge due to increasing cyber threats and data breaches. Traditional security mechanisms such as simple encryption are often insufficient to ensure complete data confidentiality and integrity. To overcome these limitations, this project titled "Digital Trapdoor: Multi-layer Security using Encryption & Steganography" introduces a combined approach that enhances data protection through multiple layers of security.

The project integrates cryptography and steganography — two powerful techniques for securing information. Cryptography converts the original data into an unreadable cipher text using encryption algorithms, ensuring that unauthorized users cannot access or interpret it. Steganography, on the other hand, conceals the encrypted data within digital media such as images, making the very existence of the hidden message undetectable to intruders. This dual-layer mechanism ensures that even if one layer is compromised, the other continues to safeguard the data.

1. **Encryption Layer:** This layer employs standard encryption algorithms (such as AES or RSA) to transform plain text into cipher text, protecting the content from unauthorized access.
2. **Steganography Layer:** In this stage, the encrypted data is embedded into a cover medium (like an image or audio file) using Least Significant Bit (LSB) or other suitable techniques, thereby achieving covert communication.
3. **Decryption and Extraction:** At the receiver's end, the hidden data is extracted and decrypted using the appropriate keys to retrieve the original information.

Overall, Digital Trapdoor highlights the importance of layered defense mechanisms in cybersecurity and showcases an innovative method to enhance the resilience of digital communication systems.

2.2. Background and Motivation

In the modern digital world, the exchange of information over the internet has become an essential part of daily life. However, this increasing reliance on online communication also brings a rise in cyber threats such as hacking, identity theft, and data breaches. Sensitive data like personal details, financial information, and confidential messages are often at risk of being intercepted or misused by unauthorized individuals.

Traditional security measures, such as basic encryption, provide a level of protection but can still be vulnerable to modern cryptanalysis and network attacks. This limitation inspired the concept of developing a multi-layer security mechanism that goes beyond encryption alone.

The project “Digital Trapdoor: Multi-layer Security using Encryption & Steganography” aims to strengthen data protection by combining two security techniques—encryption and steganography. Encryption ensures that data remains unreadable to outsiders, while steganography conceals the encrypted data within another medium, making it invisible to attackers.

This layered approach increases confidentiality and minimizes the risk of data exposure, motivating its implementation as a secure communication solution.

2.3 Objective

The primary objective of the project “**Digital Trapdoor: Multi-layer Security using Encryption & Steganography**” is to develop a secure framework that ensures the safe transmission and storage of sensitive information. The system integrates two powerful security mechanisms—encryption and steganography—to protect data from unauthorized access and cyber threats.

In the first layer, the data is encrypted using cryptographic algorithms, converting it into an unreadable format that cannot be interpreted without the correct decryption key. In the second layer, this encrypted data is embedded into a digital medium such as an image or audio file through steganographic techniques, making its presence invisible to attackers.

This dual-layered approach enhances data confidentiality, integrity, and authenticity during storage or communication. By combining encryption and steganography, the project aims to offer an efficient and reliable method for secure data exchange in modern digital environments.

2.4. Methodology

The methodology for the project “Digital Trapdoor: Multi-layer Security using Encryption & Steganography” involves a systematic approach divided into six key phases: Requirement Analysis, System Design, Implementation, Testing & Evaluation, Result Analysis, and Conclusion. Each phase plays a crucial role in ensuring that the developed system is secure, efficient, and reliable.

1. Requirement Analysis

In this phase, the need for implementing multi-layer security in digital communication is identified.

A detailed study of various encryption (e.g., AES, RSA) and steganography (e.g., LSB, DCT, DWT) algorithms is conducted to determine the most appropriate combination for secure data transmission.

System requirements are defined, including:

- Input data types (text, files, images, etc.)
- Cover media formats (image, audio, or video)
- Performance parameters such as capacity, imperceptibility, robustness, and computational efficiency.

This phase ensures a clear understanding of both functional and non-functional system requirements before proceeding to design.

2. System Design

The system architecture is designed to integrate encryption and steganography as two interdependent layers of defense.

The process is divided into two stages:

1. Encryption Layer:

The plaintext data is first encrypted using either a symmetric algorithm (AES) for

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speed or an asymmetric algorithm (RSA) for stronger key management. The output is cipher text, which is unreadable to unauthorized users.

2. Steganography

The encrypted data is then concealed within a cover medium (such as an image or audio file) using techniques like Least Significant Bit (LSB) or DCT-based embedding. This step ensures that even if the file is intercepted, the presence of hidden information remains undetectable.

Additionally, the design specifies a decryption and extraction mechanism for authorized users to retrieve and decrypt the hidden message securely.

The architecture emphasizes confidentiality, integrity, and non-detectability.

3. Implementation

The system is implemented in a modular format to simplify testing and maintenance.

- The encryption module is developed using selected cryptographic algorithms to convert plain data into cipher text.
- The steganography module embeds this cipher text into the chosen cover medium while maintaining imperceptibility.

4. Testing and Evaluation

Comprehensive testing is conducted to evaluate the performance of the system across various file types and sizes.

The evaluation parameters include:

- Security: Ability to resist unauthorized access, brute-force attacks, or detection attempts.
- Imperceptibility: The embedded data should cause minimal or no visible/audible distortion in the cover media.
- Capacity: The amount of secret data that can be hidden without compromising the cover medium's quality.

5. Result Analysis

This phase involves analyzing the system's performance and comparing it with traditional single-layer security models.

Key observations include:

- A noticeable improvement in data confidentiality and protection against steganalysis attacks due to the dual-layer approach.
- Enhanced security metrics demonstrating that combining encryption and steganography offers stronger resistance against unauthorized access. Visual, statistical, and computational analyses are performed to verify that the system meets the predefined performance criteria.

3. Hardware and Software Requirement

Hardware Requirement:

Processor	Quad-core or higher (64-bit)
RAM	Minimum 8 GB (16 GB recommended)
Storage	100 GB free space

Software Requirement:

Operating System	Ubuntu Server 20.04 LTS / Windows 10 / Kali Linux
Virtualization	VirtualBox or VMware (optional)

4. Functional Specification

1. Functional Overview

The system performs a two-step security process:

1. Encryption: Converts the original message (plaintext) into a secure, unreadable format (ciphertext) using cryptographic algorithms.
2. Steganography: Embeds the encrypted data into a digital cover medium (such as an image or audio file) using steganographic techniques, ensuring the hidden data remains imperceptible to unauthorized users.

2. Major Functional Modules

a) User Interface Module

- Provides an easy-to-use interface for performing encryption, embedding, extraction, and decryption operations.
- Allows users to select files, enter secret keys, and view process status.

b) Encryption Module

- Accepts plaintext input from the user.
- Encrypts the input using a chosen algorithm (e.g., AES or RSA).
- Generates secure ciphertext to be used for hiding.

c) Steganography Module

- Takes the ciphertext as input and embeds it into a cover medium (image, audio, or video file).
- Uses techniques like Least Significant Bit (LSB) or Discrete Cosine Transform (DCT) for data hiding.
- Produces a stego file that appears identical to the original cover medium.

d) Extraction Module

- Extracts the hidden ciphertext from the stego file using the reverse steganography process.
- Ensures accurate retrieval without altering the cover medium significantly.

e) Decryption Module

- Decrypts the extracted ciphertext using the same encryption key or algorithm.
- Recovers the original plaintext message for authorized users.

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f) Security & Validation Module

- Validates user credentials and encryption keys.
- Ensures that only authorized users can access, extract, or decrypt hidden data.

3. Input Specifications

- Plaintext data (text or file).
- Cover media (image/audio file).
- Secret encryption key.

4. Output Specifications

- Ciphertext: Encrypted version of the original data.
- Stego File: Media file containing hidden encrypted data.
- Extracted Data: Retrieved encrypted data from stego file.
- Decrypted Message: Original message after decryption.

5. Tools Description

<u>Tools & Module</u>	<u>Description (Short & Simple)</u>
OpenSSL	A command-line tool used for encryption, decryption, key generation, and SSL/TLS certificate management.
dd	A Linux command used to copy, convert, or clone data from one file or device to another (often used for disk imaging).
split	A Linux command that splits large files into smaller parts for easier storage or transfer.
Open Puff	A steganography tool that hides files secretly inside other carrier files (like images, audio, or video).
CyberChef	A web-based tool for data encryption, encoding, decoding, and analysis through drag-and-drop “recipes.”
WeTransfer	An online file-sharing service used to upload and share large files via download links.

How To Main User Create

Ubuntu main server setup – A fresh Ubuntu server was installed and configured to act as the final destination for the project.

Main user creation – A primary user named officeuser777 was created.

Password assignment – A secure password was set for this user.

Home directory setup – The system automatically generated a dedicated home directory: /home/officeuser777. Group allocation – The user was assigned to appropriate groups for controlled access.

Access controls applied – Permissions were configured to limit unauthorized access.

Prepared for attacker reach – This user acts as the endpoint where an attacker is intended to arrive for project analysis.

```
Ubuntu@ubuntu-server:~$ sudo adduser officeuser777
[sudo] password for Ubuntu:
Sorry, try again.
[sudo] password for Ubuntu:
Sorry, try again.
[sudo] password for Ubuntu:
info: Adding user `officeuser777' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `officeuser777' (1004) ...
info: Adding new user `officeuser777' (1004) with group `officeuser777 (1004)' ...
info: Creating home directory `/home/officeuser777' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for officeuser777
Enter the new value, or press ENTER for the default
    Full Name []: VADIL
    Room Number []: 01
    Work Phone []: 9685741425
    Home Phone []: 1425369685
    Other []: 0
Is the information correct? [Y/n] Y
info: Adding new user `officeuser777' to supplemental / extra groups `users' ...
info: Adding user `officeuser777' to group `users' ...
```

SSH Service

Step 1: Check if SSH is installed

Command: `sudo systemctl status ssh`

If the service is not found, you need to install SSH.

Step 2: Install OpenSSH Server

Update packages: `sudo apt update`

Install SSH: `sudo apt install openssh-server -y`

Step 3: Start and enable SSH

Start the SSH service: `sudo systemctl start ssh`

Enable SSH to start on boot: `sudo systemctl enable ssh`

Verify the status: `sudo systemctl status ssh`

```
Ubuntu@ubuntu-server:~$ sudo systemctl start ssh
Ubuntu@ubuntu-server:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable ssh
Created symlink /etc/systemd/system/ssh.service → /usr/lib/systemd/system/ssh.service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /usr/lib/systemd/system/ssh.service.
Ubuntu@ubuntu-server:~$ sudo systemctl status ssh
✦ ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Main PID: 2620 (sshd)
     Tasks: 1 (limit: 4605)
    Memory: 1.2M (peak: 1.6M)
       CPU: 27ms
   CGroup: /system.slice/ssh.service
           └─2620 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 04 05:33:50 ubuntu-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 04 05:33:50 ubuntu-server sshd[2620]: Server listening on 0.0.0.0 port 22.
Sep 04 05:33:50 ubuntu-server sshd[2620]: Server listening on :: port 22.
Sep 04 05:33:50 ubuntu-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Normal User

An Ubuntu main server was set up as the final destination where the attacker is intended to reach.

Two normal users with limited privileges were created as decoys, each with secure passwords and dedicated home directories.

These users acted as a distraction, making the attacker spend more time on irrelevant accounts before reaching the actual target

```
Ubuntu@ubuntu-server:~$ sudo adduser employee01
info: Adding user `employee01' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `employee01' (1005) ...
info: Adding new user `employee01' (1005) with group `employee01 (1005)' ...
info: Creating home directory `/home/employee01' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
Sorry, passwords do not match.
passwd: Authentication token manipulation error
passwd: password unchanged
Try again? [y/N] Y
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for employee01
Enter the new value, or press ENTER for the default
  Full Name []: CHOTA
  Room Number []: 02
  Work Phone []: 7412589656
  Home Phone []: 1475754154
  Other []: 0
Is the information correct? [Y/n] Y
info: Adding new user `employee01' to supplemental / extra groups `users' ...
info: Adding user `employee01' to group `users' ...
```

```
Ubuntu@ubuntu-server:~$ sudo adduser employee02
info: Adding user `employee02' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `employee02' (1006) ...
info: Adding new user `employee02' (1006) with group `employee02 (1006)' ...
info: Creating home directory `/home/employee02' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for employee02
Enter the new value, or press ENTER for the default
  Full Name []: OPPQ
  Room Number []: 03
  Work Phone []: 0104052014
  Home Phone []: 1010011010
  Other []: 0
Is the information correct? [Y/n] Y
info: Adding new user `employee02' to supplemental / extra groups `users' ...
info: Adding user `employee02' to group `users' ...
Ubuntu@ubuntu-server:~$
```

Encryption

```
(kali@kali)-[~/Desktop/Ubuntu]  
$ openssl enc -aes-256-cbc -pbkdf2 -iter 100000 -salt -in password.txt -out encrypted_password.bin
```

OpenSSL is a robust toolkit for implementing SSL and TLS protocols and a general-purpose cryptography library. It is used for encrypting data, generating and managing keys and certificates, creating digital signatures, and securing communications over networks

The password file was encrypted using AES-256-CBC encryption with PBKDF2 key derivation and 100000 iterations to ensure strong protection. The original file password.txt was converted into an encrypted file named encrypted_password.bin. A secure passphrase was applied to enhance the encryption strength and prevent unauthorized access.



Partition

```
(kali㉿kali)-[~/Desktop/Ubuntu]
$ dd if=encrypted_password.bin of=part_aa bs=10 count=1
dd if=encrypted_password.bin of=part_ad bs=10 skip=1
dd if=encrypted_password.bin of=part_ae bs=10 skip=2

1+0 records in
1+0 records out
10 bytes copied, 0.000844584 s, 11.8 kB/s
2+1 records in
2+1 records out
22 bytes copied, 4.2836e-05 s, 514 kB/s
1+1 records in
1+1 records out
12 bytes copied, 0.000137877 s, 87.0 kB/s
```

```
(kali㉿kali)-[~/Desktop/Ubuntu]
$ split --bytes=100 encrypted_password.bin part_

(kali㉿kali)-[~/Desktop/Ubuntu]
$ dd if=encrypted_password.bin of=part_aa bs=10 count=1
dd if=encrypted_password.bin of=part_ab bs=10 skip=1
dd if=encrypted_password.bin of=part_ac bs=10 skip=2

1+0 records in
1+0 records out
10 bytes copied, 0.000225301 s, 44.4 kB/s
2+1 records in
2+1 records out
```

First, you encrypted the file and generated encrypted_password.bin.

Then you used the split command to divide the file into 100-byte chunks, which created files like part_aa, part_ab, etc.

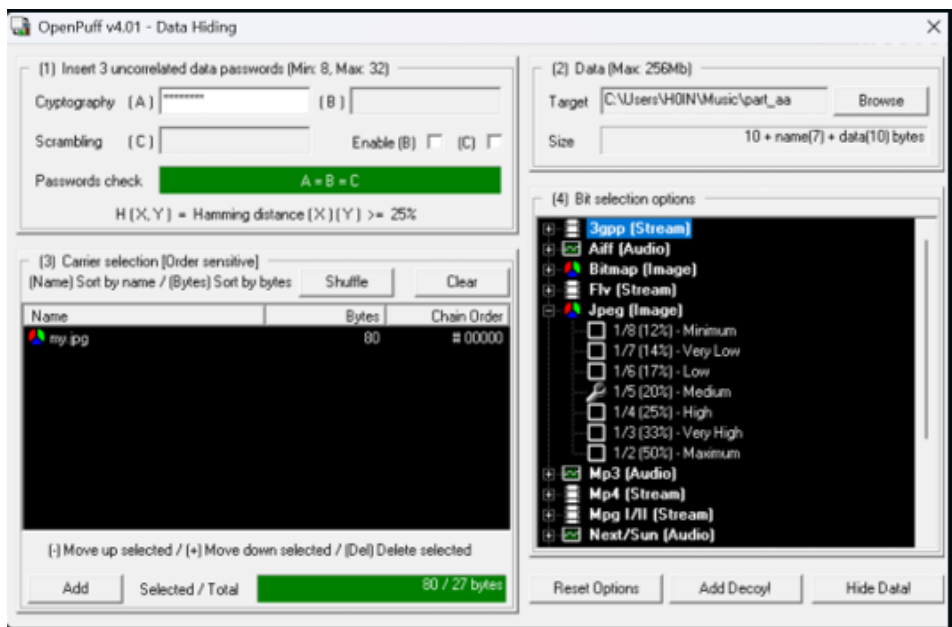
After that, you took the same encrypted file and used the dd command to extract specific portions.

With dd you divided the file further into 5 smaller parts by specifying block size (bs) and using count or skip options.

Now you have multiple smaller segments of the original encrypted file, created in two ways: first by split into bigger chunks, and second by dd into finer 10-byte pieces

Steganography.

Steganography of Image using Open Puff



Selected an image file as the carrier.

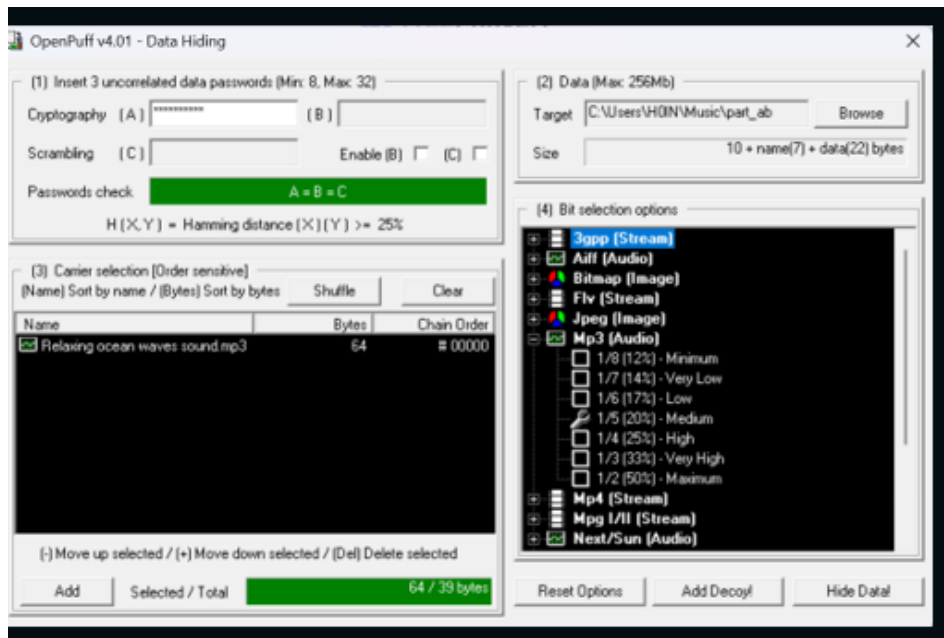
Loaded the part_aa partition (a fragment of the encrypted password).

Used OpenPuff to hide the file within the image using secure steganographic encoding.

The image appears normal but secretly contains the hidden data.

This step adds a layer of deception in the Digital Trapdoor process.

Steganography of Audio using Open Puff



Selected an audio file as the carrier.

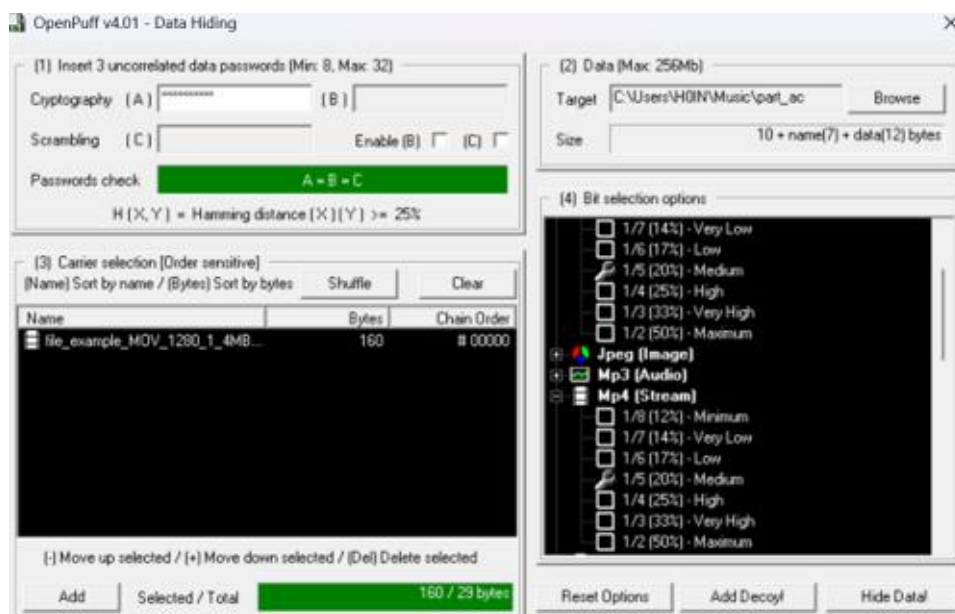
Loaded the part_ab partition (a fragment of the encrypted password).

Used OpenPuff to embed the file securely within the audio.

The audio plays normally but secretly holds the hidden data.

This step strengthens the multi-layered deception in the Digital Trapdoor process.

Steganography of Video using Open Puff



Selected a video file as the carrier.

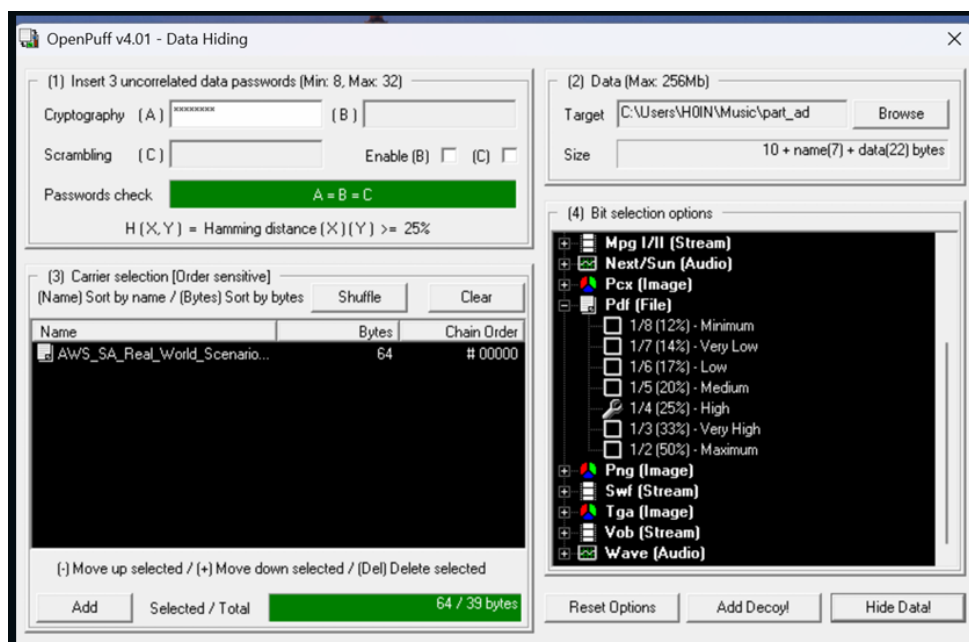
Loaded the part_ac partition (a fragment of the encrypted password).

Used OpenPuff to embed the file securely within the video.

The video plays normally but secretly contain the hidden data.

This step adds another layer of complexity in the Digital Trapdoor process.

Steganography of Text using Open Puff



Selected a text file as the carrier.

Loaded the part_ad partition (a fragment of the encrypted password).

Used OpenPuff to embed the file securely within the text file.

The text appears normal but secretly contains the hidden data.

This step continues the multi-layered deception in the Digital Trapdoor process.

Cyber Chef

1. Extract 5 Keys from Stego Files

After hiding your partitioned/encrypted parts across different files (images, audio, logs, etc.) using steganography, you now have 5 keys (these could be passphrases, secret tokens, or hash strings).

2.Open CyberChef

Go to CyberChef (online tool) or run it locally.

CyberChef allows you to apply various operations (encoding, decoding, encryption, hashing, etc.) on data.

3.Input Keys into CyberChef

Copy each key one by one into CyberChef's input window.

4. Select Operation → "To Base62"

Drag the "To Base62" operation into the "Recipe" section.

This converts your plain key into a Base62 encoded form.

Base62 uses characters [A–Z, a–z, 0–9] → making the key look random but still reversible.

5.Execute & Generate Encrypted Keys

Run the recipe.

Each key will now be converted to Base62 encoded text.

Example: Original key → mySecret123

Base62 encoded → 1AbX9Kjz... (random-looking string).

6.Repeat for All 5 Keys

Perform Base62 encoding for all 5 keys separately.

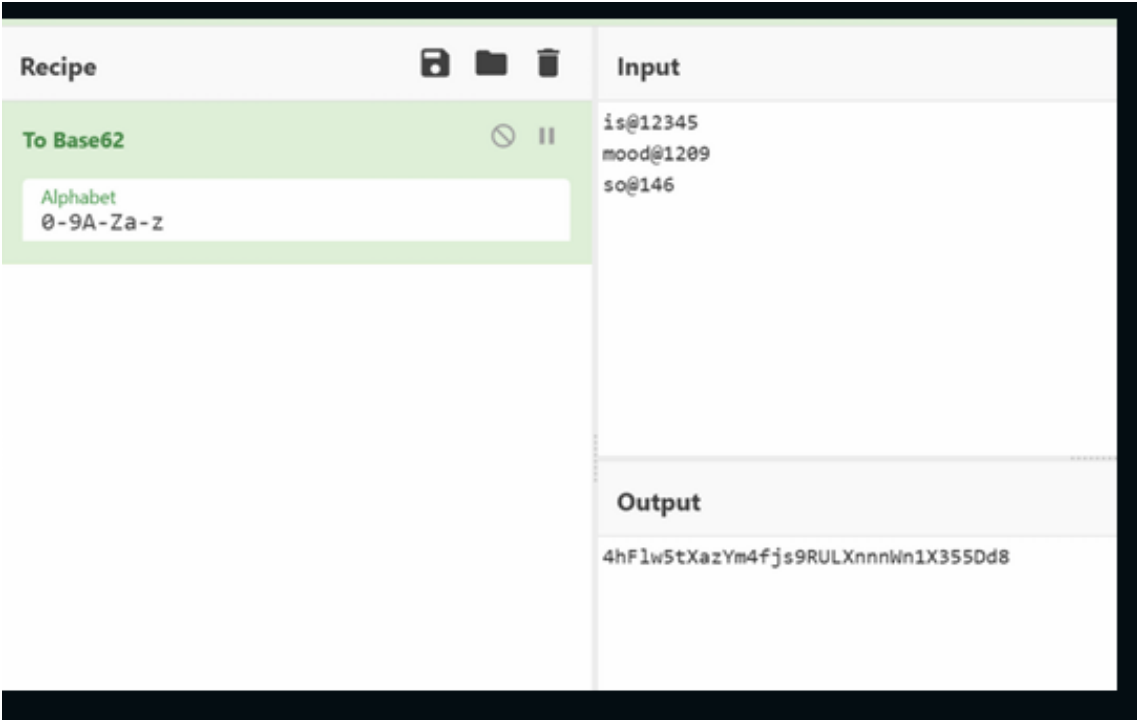
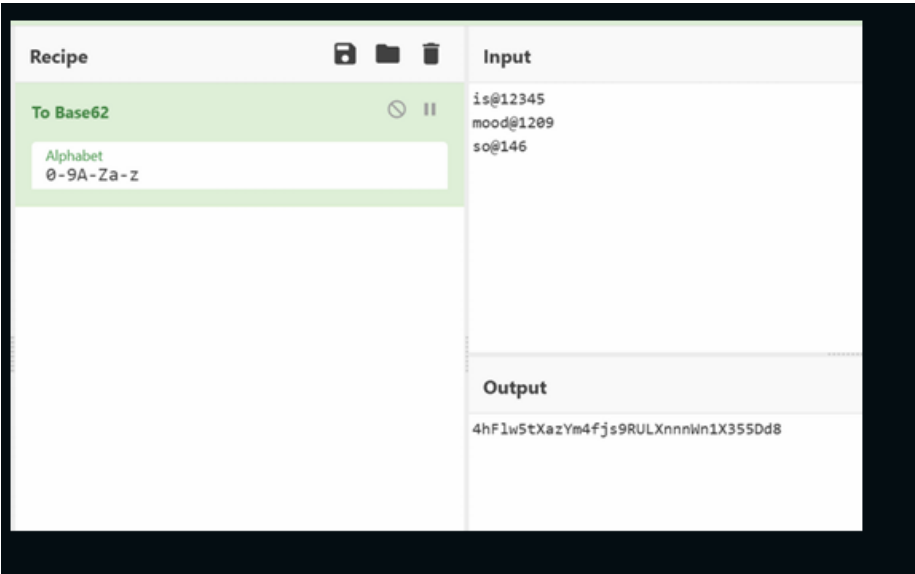
You'll get 5 different Base62 strings.

7.Store Encoded Keys in a TXT File

Open a new text file (e.g., keys_encoded.txt).

Save each encoded key on a new line for clarity.

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We Transfer

1. Upload Encoded Files

Five Base64-encoded .txt fragments are uploaded to WeTransfer.

Each upload generates a unique public download link.

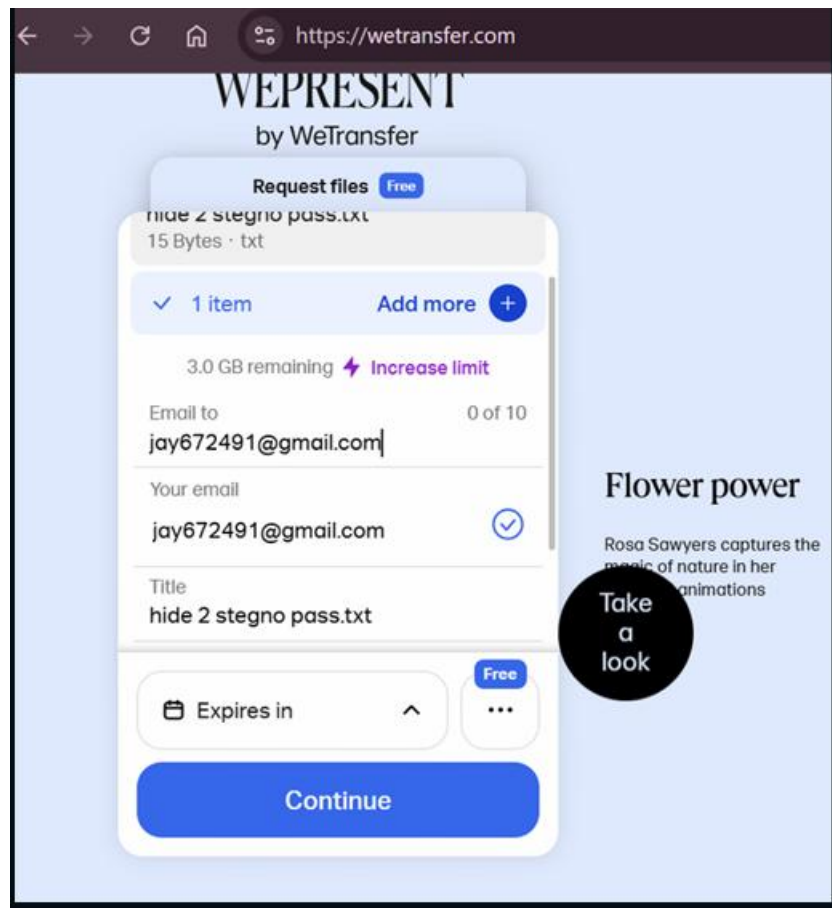
2. Collect Public Links

All five WeTransfer links are noted down.

These links are the only way to retrieve the fragments.

3. Embed Links into Logs

Each link is hidden inside log files using a fixed marker pattern



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Audit Log (audit.log) → Put the link at the end of a USER_LOGIN entry in the audit.log file. along with a marker.

3	Sep 15 07:35:30 audit[5758]: pid=5758 uid=guest logout				
9	Sep 15 07:37:30 audit[6703]: pid=6703 uid=user2 syscall=read				
0	Sep 15 07:39:30 audit[1924]: pid=1924 uid=root syscall=read				
1	Sep 15 07:41:30 audit[4433]: pid=4433 uid=guest syscall=read				
2	Sep 15 07:43:30 audit[3144]: pid=3144 uid=root logout				
3	Sep 15 07:45:30 audit[1148]: pid=1148 uid=user2 logout				
4	Sep 15 07:47:30 audit[1946]: pid=1946 uid=guest selinux enforcement enabled				
5	Sep 15 07:49:30 audit[1450]: pid=1450 uid=admin login success				
5	Sep 15 07:51:30 audit[7841]: pid=7841 uid=user2 login success				
7	Sep 15 07:53:30 audit[5757]: pid=5757 uid=user1 login success				
3	Sep 15 07:55:30 audit[8502]: pid=8502 uid=user1 sudo CMD=/bin/l				
9	Sep 15 07:57:30 audit[5202]: pid=5202 uid=user1 sudo CMD=/bin/l				
0	Sep 15 07:59:30 audit[8987]: pid=8987 uid=admin accessed /etc/				
1	Sep 15 08:01:30 audit[1916]: pid=1916 uid=user1 sudo CMD=systemctl restart ssh				
2	Sep 15 08:03:30 audit[3615]: pid=3615 uid=admin sudo CMD=systemctl restart ssh				

System Log (syslog) → In the syslog file, add the link at the end of a system services start/stop message, along with a marker.

Sep 15 08:25:00 host2 sshd[3961]: Failed password for invalid user admin from 192.168.1.100 port 3271					
Sep 15 08:30:00 host2 apt[6239]: Installed package nginx					
Sep 15 08:35:00 host2 sudo[8479]: user3 : TTY=pts/3 ; PWD=/home/user ; COMMAND=/bin/l					
Sep 15 08:40:00 host2 systemd[1315]: Started Session					
Sep 15 08:45:00 host2 sudo[6176]: user2 : TTY=pts/4 ; PWD=/home/user ; COMMAND=/bin/l					
Sep 15 08:50:00 host2 apt[3118]: Installed package nginx					
Sep 15 08:55:00 host2 sshd[4127]: Accepted password for user4 from 10.0.0.32 port 45319 ssh2					
Sep 15 09:00:00 host2 NetworkManager[5300]: Device eth0 connected					
Sep 15 09:05:00 host2 cron[8893]: (root) CMD (backup.sh)					
Sep 15 09:10:00 host2 NetworkManager[3258]: Device eth1 connected					
Sep 15 09:15:00 host2 NetworkManager[2579]: Device eth0 connected					
Sep 15 09:20:00 host2 kernel[7155]: [INFO] USB devi					
Sep 15 09:25:00 host2 sudo[8634]: user3 : TTY=pts/2 ; PWD=/home/user ; COMMAND=/bin/l					
Sep 15 09:30:00 host2 cron[6926]: (root) CMD (logrotate)					
Sep 15 09:35:00 host2 systemd[178]: Started Session					

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

Firewall Log (ufw.log) → Add the link at the end of a UFW BLOCK/ALLOW rule, along with a marker.

```
Sep 15 13:51:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.8 DST=malicious.example.com LEN=100 PROTO=TCP SPT=60707 DPT=4444
Sep 15 13:52:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.6 DST=malicious.example.com LEN=100 PROTO=TCP SPT=62080 DPT=4444
Sep 15 13:53:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.6 DST=malicious.example.com LEN=100 PROTO=TCP SPT=41027 DPT=4444
Sep 15 13:52:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.7 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=49470 DPT=8080 RST
Sep 15 13:53:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.8 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=43773 DPT=8080 RST
Sep 15 13:54:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.5 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=48386 DPT=8080 RST
Sep 15 13:55:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.5 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=57813 DPT=8080 RST
Sep 15 13:56:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.5 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=62871 DPT=8080 RST
Sep 15 13:57:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.6 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=58813 DPT=8080 RST
Sep 15 13:58:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.8 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=48946 DPT=8080 RST
Sep 15 13:59:00 host-fw kernel: [UFW BLOCK] OUT=eth0 SRC=10.0.0.6 DST=185.244.12.7 LEN=100 PROTO=TCP SPT=63439 DPT=8080 RST
Sep 15 14:20:00 host-fw kernel: [UFW ALLOW] OUT=eth0 SRC=10.0.0.5 DST=8.8.8.8 PROTO=UDP SPT=53535 DPT=53 LEN=70
Sep 15 14:22:00 host-fw kernel: [UFW ALLOW] OUT=eth0 SRC=10.0.0.5 DST=93.184.216.34 PROTO=TCP SFhttps://we.tl/t-u7qLY3Y5d8
```

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

Authentication Log (auth.log) → Add the link at the end of an SSH login success/failure entry, along with a marker.

```
Sep 15 10:55:11 server1 sshd[1245]: Failed
password : https://we.tl/t-iv7X957Nk6 for
invalid user root from 198.51.100.45 port
51234 ssh2
```

Antivirus Log (antivirus.log) → In the antivirus.log file, add the link at the end of a scan complete/threat detected entry, along with a marker.

Sep 15 07:42:30 audit[1155]: pid=1155 uid=guest system read			
Sep 15 07:43:30 audit[3144]: pid=3144 uid=root logout			
Sep 15 07:45:30 audit[1148]: pid=1148 uid=user2 logout			
Sep 15 07:47:30 audit[1946]: pid=1946 uid=guest selinux enforcement enabled			
Sep 15 07:49:30 audit[1450]: pid=1450 uid=admin login success			
Sep 15 07:51:30 audit[7841]: pid=7841 uid=user2 login success			
Sep 15 07:53:30 audit[5757]: pid=5757 uid=user1 login success			
Sep 15 07:55:30 audit[8502]: pid=8502 uid=user1 sudo CMD=/bin/l			
Sep 15 07:57:30 audit[5202]: pid=5202 uid=user1 sudo CMD=/bin/l			
Sep 15 07:59:30 audit[8987]: pid=8987 uid=admin accessed /et	https://we.tl/t-NFpsU24grV		
Sep 15 08:01:30 audit[1916]: pid=1916 uid=user1 sudo CMD=systemctl restart ssh			

Winscp.

WinSCP is primarily used to transfer files securely between a Windows PC and remote systems like Linux/Unix servers using encrypted protocols (SFTP or SCP). It also includes a graphical user interface (GUI), command-line interface, and scripting support for automation.



1. I used WinSCP, a secure file transfer client, to share files from my Windows system to an Ubuntu server using the SFTP (Secure File Transfer Protocol).
2. In the WinSCP interface, I entered the IP address of the Ubuntu system where the files needed to be uploaded, along with the username and its corresponding password to establish a secure connection.
3. Once connected, I was able to upload files directly from my local Windows machine to the remote Ubuntu server securely and efficiently. This method ensures data integrity and encryption during transfer, providing a safe and reliable way to exchange files between the two systems.
4. After enter login

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

The image shows a 'Login' dialog box with a yellow title bar. On the left is a 'New Site' panel. The main area is titled 'Session' and contains the following fields: 'File protocol:' set to 'SFTP', 'Host name:' set to '192.168.0.126', 'Port number:' set to '22', 'User name:' set to 'employee01', and 'Password:' masked with dots. Below these are 'Save' and 'Advanced...' buttons. At the bottom are 'Tools' and 'Manage' dropdowns, a 'Login' button with a green icon, 'Close', and 'Help' buttons. A checkbox at the bottom left is checked and labeled 'Show Login dialog on startup and when the last session is closed'.

Session

File protocol:
SFTP

Host name: 192.168.0.126 Port number: 22

User name: employee01 Password:

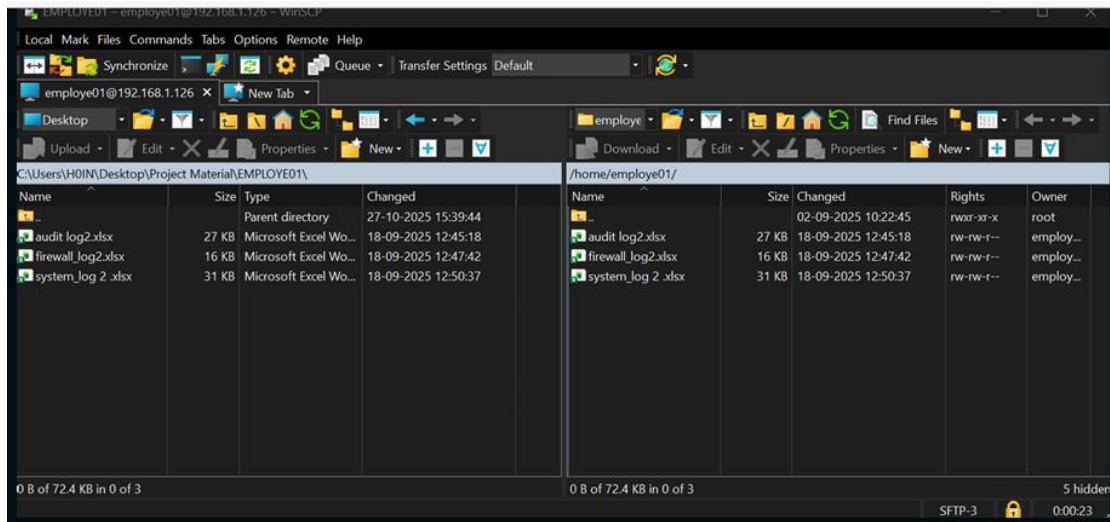
Save Advanced...

Tools Manage Login Close Help

☒ Show Login dialog on startup and when the last session is closed

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

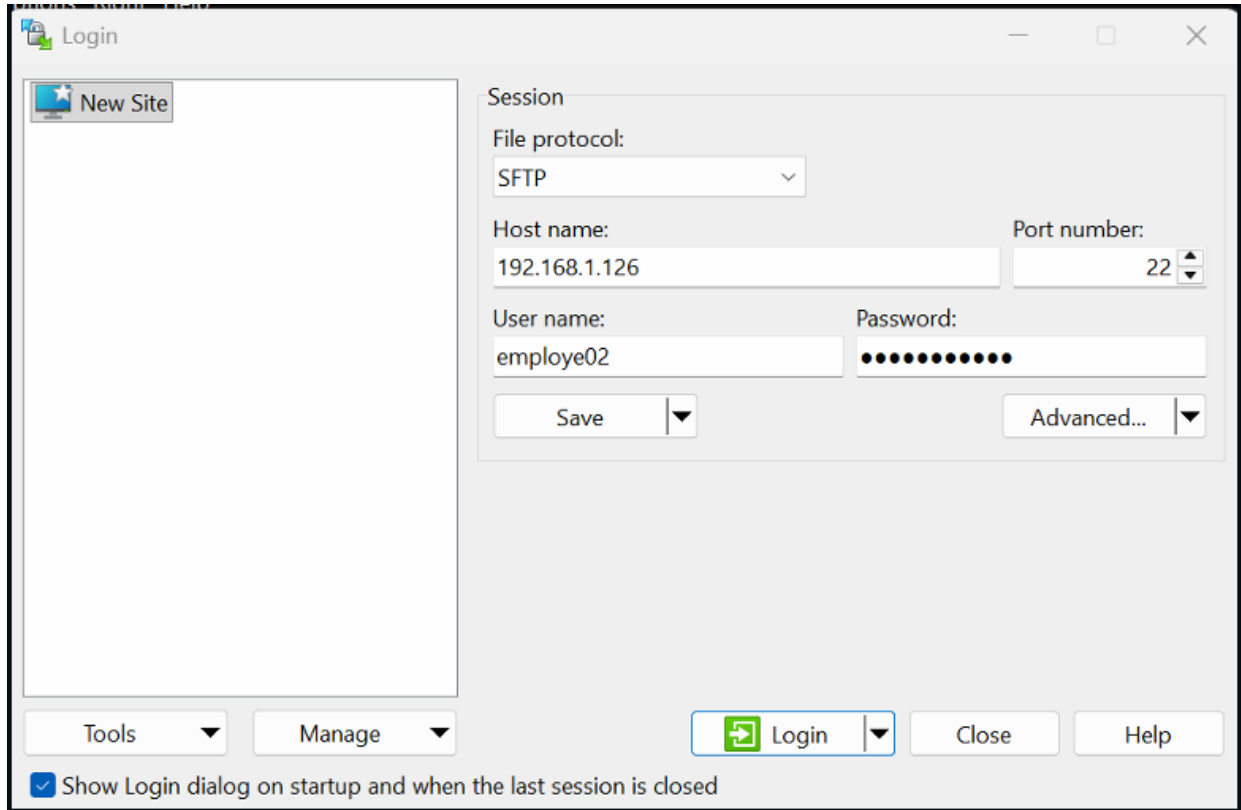
Once you select the file and click Upload, it will be transferred to the Ubuntu system.



```
employee01@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:72:9f:e0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.126/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 7141sec preferred_lft 7141sec
    inet6 fe80::a00:27ff:fe72:9fe0/64 scope link
        valid_lft forever preferred_lft forever
employee01@ubuntu-server:~$ ls -a
.  ..  'audit log2.xlsx'  .bash_history  .bash_logout  .bashrc  .cache  firewall_log2.xlsx  .profil
employee01@ubuntu-server:~$
```

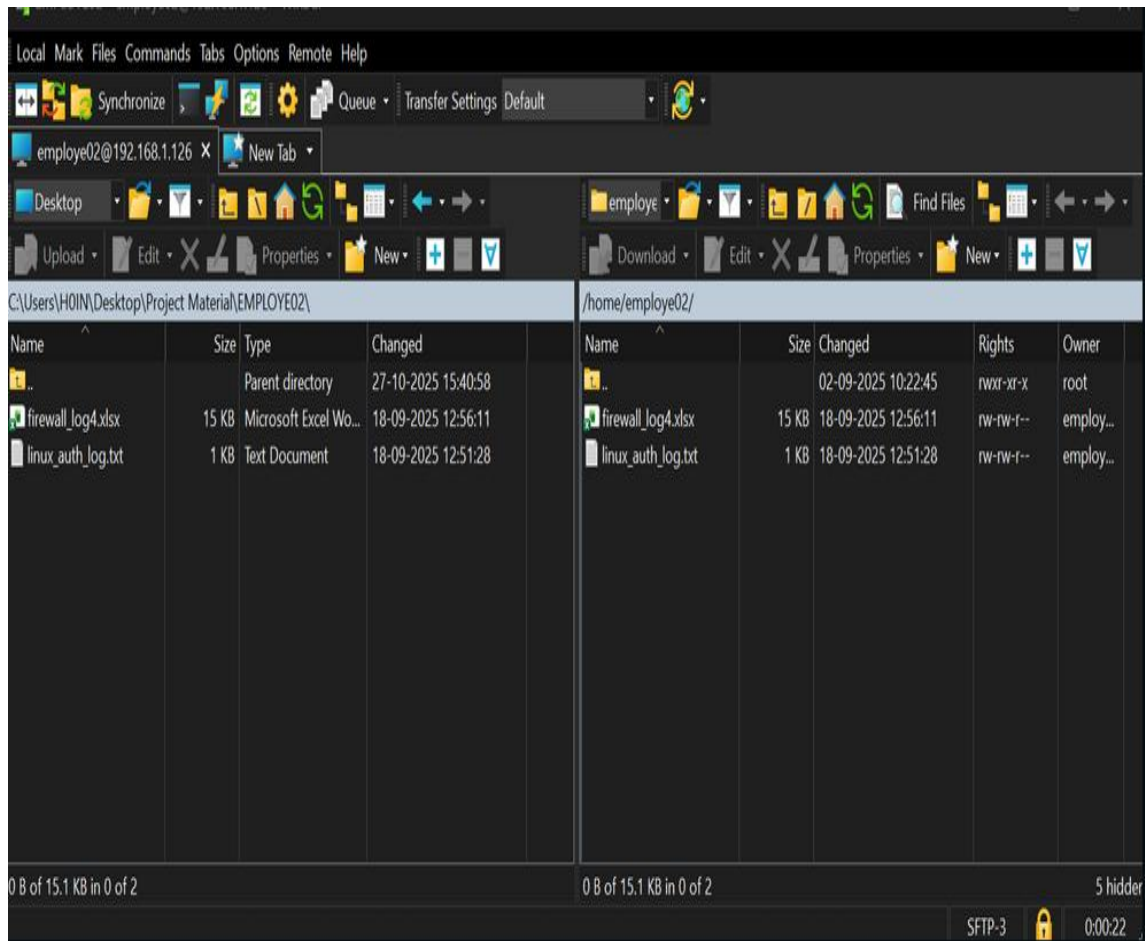
Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

In the WinSCP interface, I entered the IP address of the Ubuntu system where the files needed to be uploaded, along with the username and its corresponding password to establish a secure connection.



Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

Once you select the file and click Upload, it will be transferred to the Ubuntu system.



```
employee02@ubuntu-server:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:72:9f:e0 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.126/24 metric 100 brd 192.168.1.255 scope global dynamic enp0s3
        valid_lft 6041sec preferred_lft 6041sec
    inet6 fe80::a00:27ff:fe72:9fe0/64 scope link
        valid_lft forever preferred_lft forever
employee02@ubuntu-server:~$ ls -la
.  ..  .bash_history  .bash_logout  .bashrc  .cache  firewall_log4.xlsx  linux_auth_log.txt  .profile
```

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

After we uploaded the log file to the Ubuntu server, we gave the attacker the user details so they could perform a brute-force attack. The request is to convert those user details into an encrypted/encoded format using CyberChef.

```
Name: ketan-emp Verma
Phone: +91-70000-00001
Email: rohan.ployee.test@example.com

Address: 12 Test Park Lane, Mocktown
Pincode: 560000
Credit card: **** * 0001
```

```
50PZÉOUêæhøhPî%¼40M40MDP¥PZPhÉc-
zÈ^Å00ç(P]è,xdp²ÓÚ®BÚPã(rKhÂsâÊP{P´ÓM-
çbμ£«wM4Ô
```

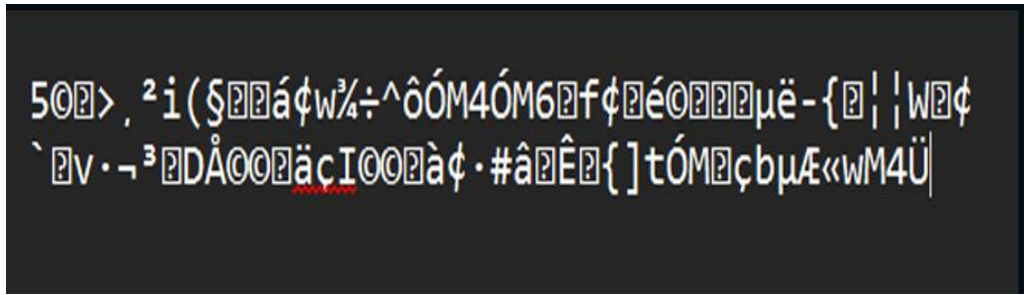
This is the encrypted data that was sent to the attacker via the link.

same as first then we forward to second user details we gave the attacker the user details so they could perform a brute-force attack. The request is to convert those user details into an encrypted/encoded format using CyberChef.

```
Name: Priya Singh
Phone: +91-70000-00002
Email: emp.loye.test@example.com

Address: 45 Example Street, Sample City
Pincode: 110000
Credit card: **** * 0002
```

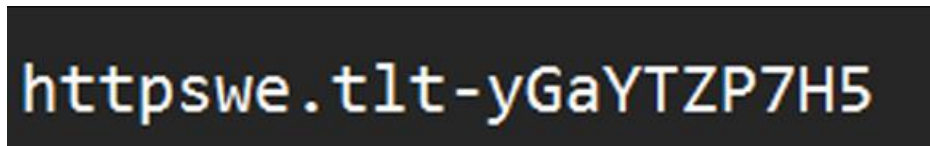
Digital Trapdoor: Multi-layer Security using Encryption & Steganography.



This is the encrypted data that was sent to the attacker via the link.

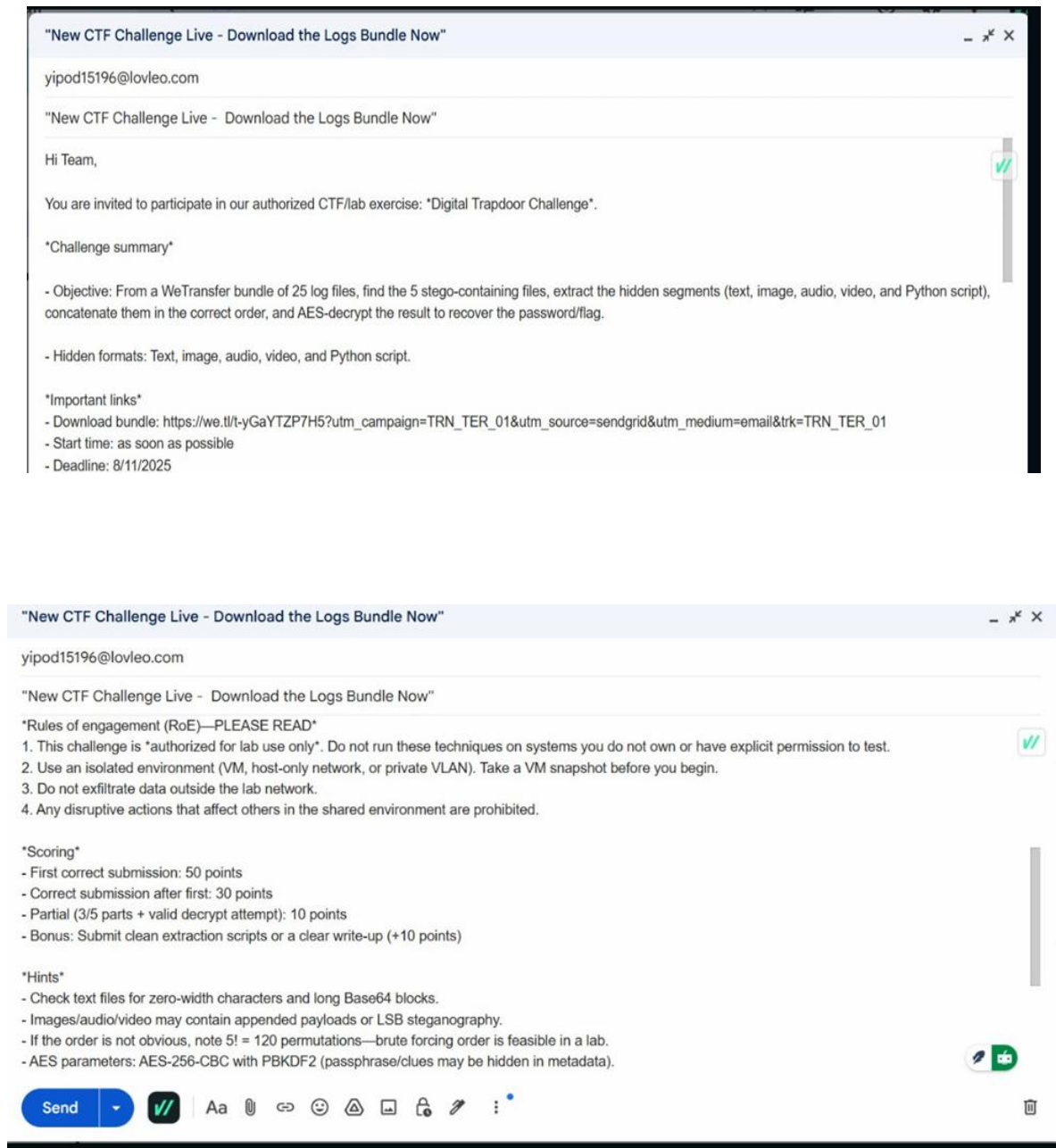
1. After uploading the log file to the Ubuntu server, we provided the attacker with a link containing the encrypted user details. The file delivered via that link contains the user credentials in encrypted form; the attacker received the ciphertext but not the decryption key. This entry documents that the data was transmitted as an encrypted payload and identifies the delivery method (a shared link) without exposing any sensitive information or decryption material.

Then send Attacker through the link.



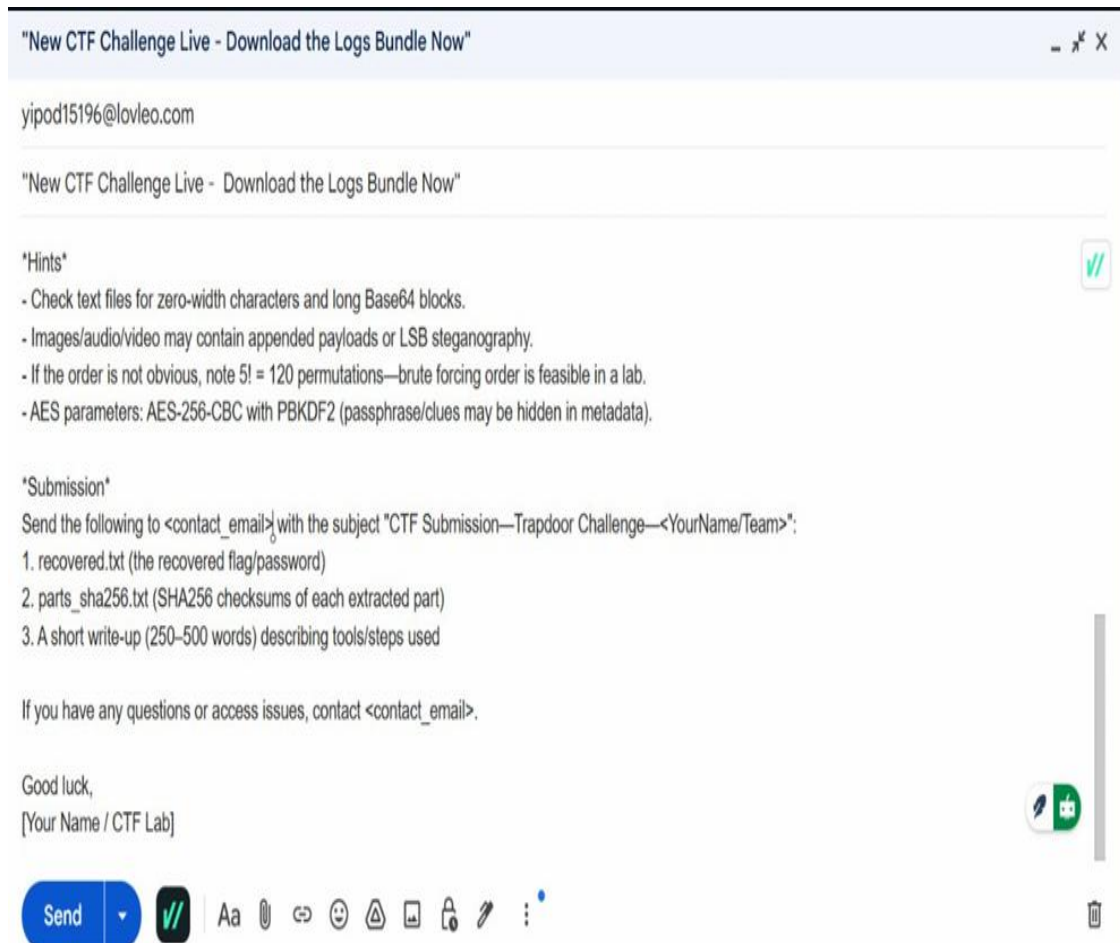
6. Proof of Concept

- Open The Mail

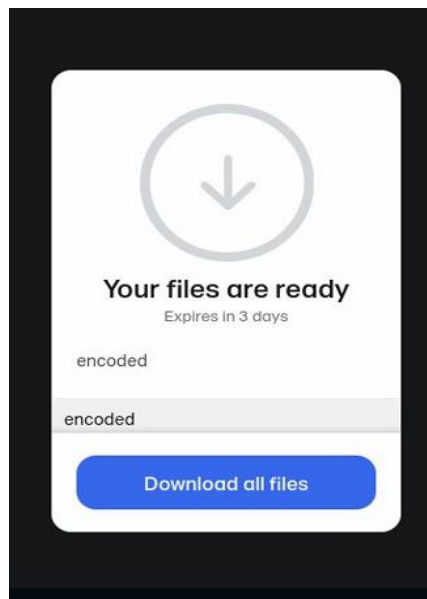


- They first opened the email and read all of its contents.

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.



- After reading, they copied the link from the message and pasted it into the browser.



Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

- On the resulting page they discovered one folder whose name is encoded into found 2 files.

Name	Size	Type	Date Modified
01.txt	149 bytes	Plain text d...	01 November 2025, 07:48
02.txt	143 bytes	Plain text d...	01 November 2025, 07:48

- On the resulting page they discovered two encoded files that contained hidden username details.

```
1 50eZéUæhøh1x%40M40MD"¥Z hËçzË^Ä©ç( ]ë,xdp²00®BÜã(rKhÂsâÊt'ÔMçbμÆ«wM4Ô
```

- After open one file they found encode text.

The screenshot shows a web-based Base64 decoder interface. On the left, under the heading "Recipe", the "To Base64" section is active, showing the "Alphabet" as "A-Za-z0-9+/=". On the right, the "Input" field contains the encoded string: `50eZéUæhøh1x%40M40MD"¥Z hËçzË^Ä©ç( ]ë,xdp²00®BÜã(rKhÂsâÊt'ÔMçbμÆ«wM4Ô`. Below the input, the "Output" field displays the decoded text: `NameketanempVermaPhone+917000000001EmailrohanployeetestexamplecomCBdress12TestParkLaneMocktownPIncode560000Credit card0001A==`.

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

- Using CyberChef they decoded those files to reveal the usernames and other relevant information.

```
3
4 Name:ketanempVerma
5 Phone+917000000001
6 Email:rohanployeetestexample.com
7 Adress:12TestParkLaneMocktown
8 Pincode:560000
9 Creditcard:0001A=
```

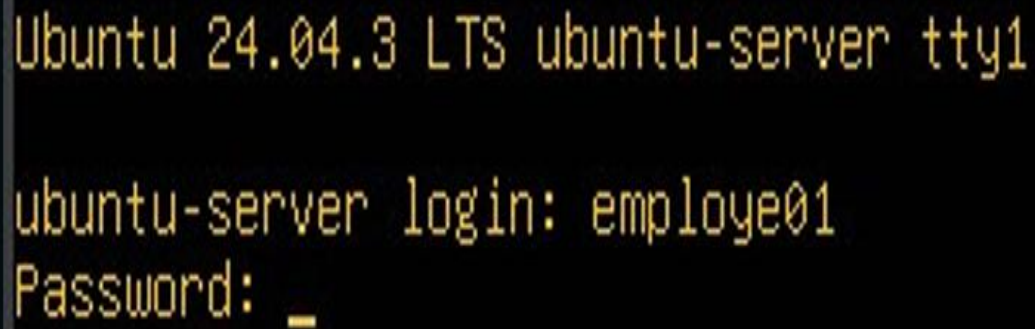
- With the collected details in hand, they constructed a password list intended for a brute- force attempt.

```
(kali@kali)-[~/Desktop]
└─$ hydra -L users.txt -P password.txt -t 4 -f -o hydra_ssh_out.txt ssh://10.75.36.32
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-11-02 11:00:04
[DATA] max 4 tasks per 1 server, overall 4 tasks, 103 login tries (l:1/p:103), ~26 tries per task
[DATA] attacking ssh://10.75.36.32:22/
[STATUS] 61.00 tries/min, 61 tries in 00:01h, 42 to do in 00:01h, 4 active
[22][ssh] host: 10.75.36.32 login: employe01 password: EMPLOYE@01#
[STATUS] attack finished for 10.75.36.32 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-11-02 11:01:47
```

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

- They launched the attack using Hydra and, after some effort, obtained a valid username and password.

A terminal window with a black background and yellow text. The first line shows the system boot message: 'Ubuntu 24.04.3 LTS ubuntu-server tty1'. The second line shows the login prompt: 'ubuntu-server login: employee01'. The third line shows the password prompt: 'Password: _', where the underscore represents a single character entered.

```
Ubuntu 24.04.3 LTS ubuntu-server tty1
ubuntu-server login: employee01
Password: _
```

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

- sing those credentials they logged into the client system and expanded the user information with getent, which revealed three files.

```
Ubuntu 24.04.3 LTS ubuntu-server tty1
ubuntu-server login: employe01
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-86-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sun Nov  2 04:04:57 PM UTC 2025

System load:          0.05
Usage of /:            15.1% of 24.44GB
Memory usage:         5%
Swap usage:           0%
Processes:            111
Users logged in:      0
IPv4 address for enp0s3: 10.75.36.32
IPv6 address for enp0s3: 2401:4900:aab3:9349:a00:27ff:fe72:9fe0

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

38 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

employe01@ubuntu-server:~$ _
```

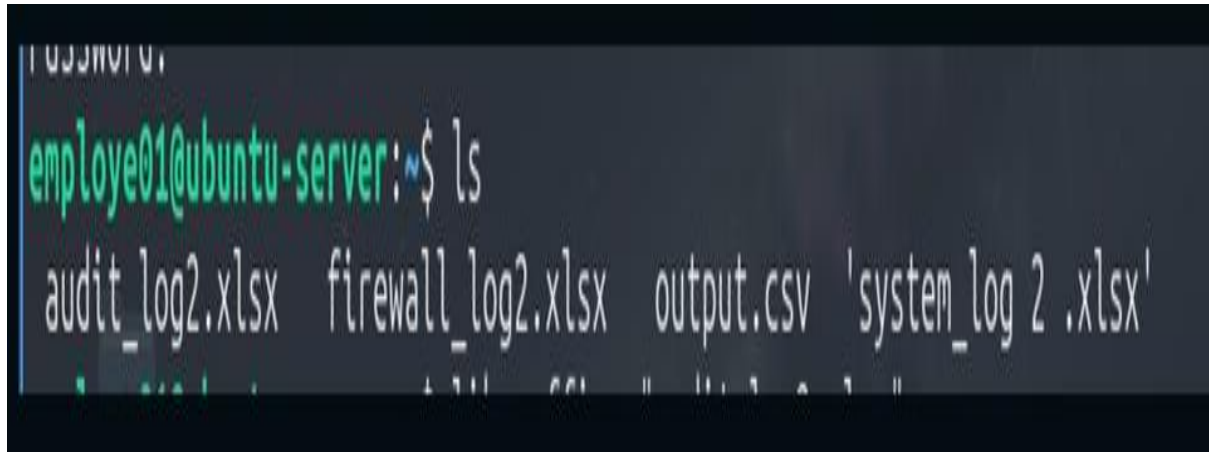
Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

```
sn
Ubuntu@ubuntu-server:~$ getent passwd employee01
employee01:x:1005:1005:CHOTA,02,7412589656,1475754154,0:/home/employee01:/bin/bash
Ubuntu@ubuntu-server:~$ getent passwd employee02
employee02:x:1006:1006:OPPO,03,0104052014,1010011010,0:/home/employee02:/bin/bash
```

- They retrieve entries from administrative databases using the getent command.
- They found employee01 and employee02 details

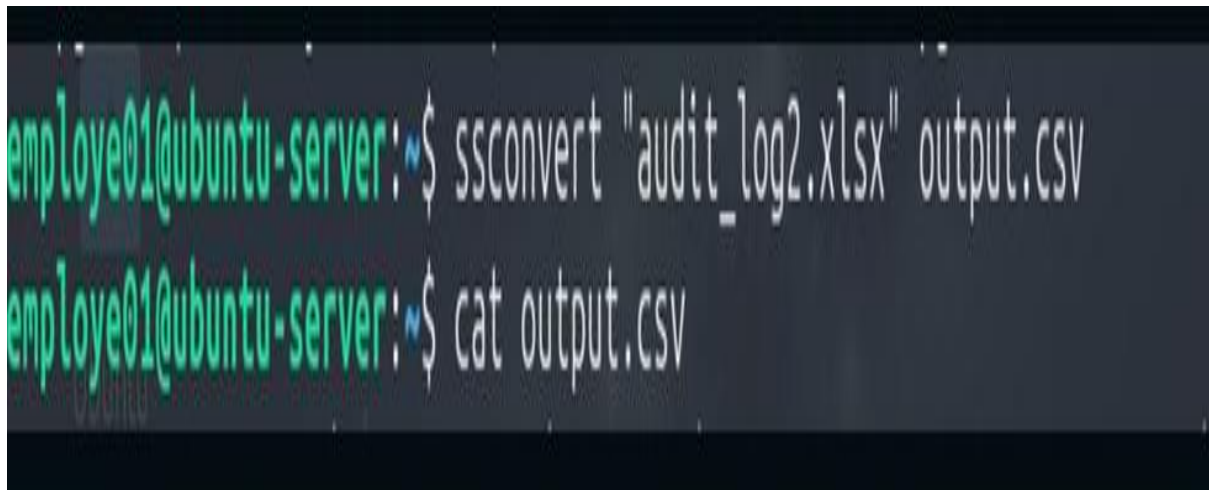
```
employee02:x:1006:1006:OPPO,03,0104052014,1010011010
Ubuntu@ubuntu-server:~$ su - employee01
Password:
employee01@ubuntu-server:~$ ls
```

- Through the normal ubuntu user they switch into employee01 user using username and password

A terminal window screenshot showing a user named 'employee01' at a machine named 'ubuntu-server'. The user has entered the 'ls' command, and the output lists four files: 'audit_log2.xlsx', 'firewall_log2.xlsx', 'output.csv', and 'system_log 2 .xlsx'.

```
employee01@ubuntu-server:~$ ls
audit_log2.xlsx  firewall_log2.xlsx  output.csv  'system_log 2 .xlsx'
```

- They explore all file using ls command they found three log files.

A terminal window screenshot showing two commands being executed. First, the user runs 'ssconvert "audit_log2.xlsx" output.csv' to convert an Excel file to CSV. Then, they run 'cat output.csv' to view the contents of the converted file.

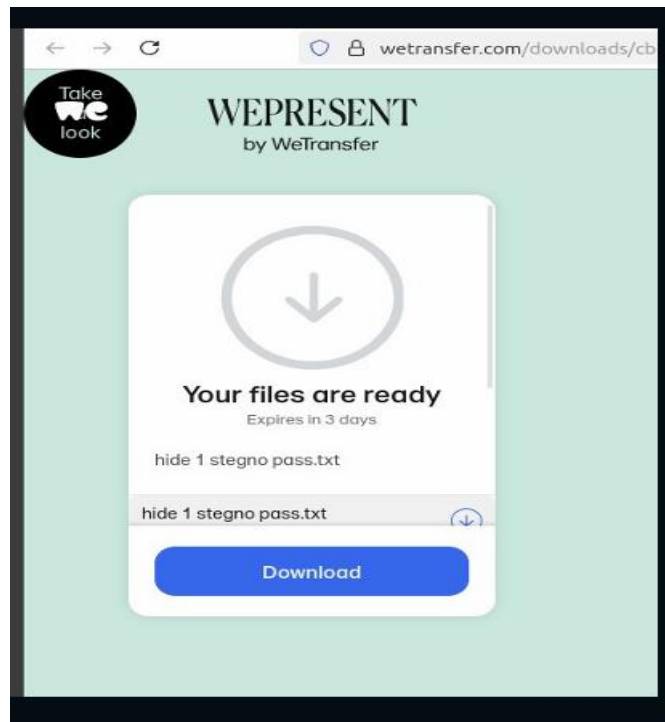
```
employee01@ubuntu-server:~$ ssconvert "audit_log2.xlsx" output.csv
employee01@ubuntu-server:~$ cat output.csv
```

- They examined the files one by one; one was an XLSX that they converted to CSV with convert and then opened for analysis.

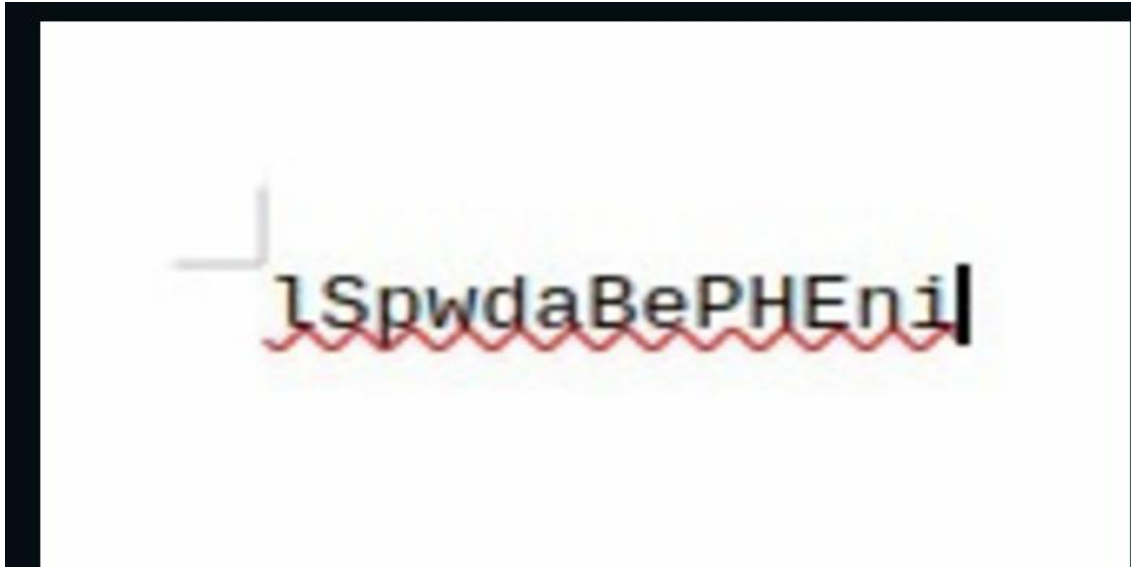
```
,
"Sep 15 07:15:30 audit[6493]: pid=6493 uid=root syscall=write",,,,,,
"Sep 15 07:17:30 audit[5921]: pid=5921 uid=guest accessed /etc/shadow",,,,,,
"Sep 15 07:19:30 audit[6585]: pid=6585 uid=root exe=/bin/cat /etc/passwd",,,,,,
"Sep 15 07:21:30 audit[4301]: pid=4301 uid=user2 login success",,,,,,
"Sep 15 07:23:30 audit[5254]: pid=5254 uid=user2 selinux enforcement enabled",,,,,,
"Sep 15 07:25:30 audit[3848]: pid=3848 uid=admin syscall=unlink",,,,,,
"Sep 15 07:27:30 audit[7204]: pid=7204 uid=user1 logout",,,,,,
"Sep 15 07:29:30 audit[4085]: pid=4085 uid=user1 login success",,,,,,
"Sep 15 07:31:30 audit[3777]: pid=3777 uid=user1 logout",,,,,,
"Sep 15 07:33:30 audit[2992]: pid=2992 uid=user2 selinux enforcement enabled",,,,,,
"Sep 15 07:35:30 audit[5758]: pid=5758 uid=guest logout",,,,,,
"Sep 15 07:37:30 audit[6703]: pid=6703 uid=user2 syscall=read",,,,,,
"Sep 15 07:39:30 audit[1924]: pid=1924 uid=root syscall=read",,,,,,
"Sep 15 07:41:30 audit[4433]: pid=4433 uid=guest syscall=read",,,,,,
"Sep 15 07:43:30 audit[3144]: pid=3144 uid=root logout",,,,,,
"Sep 15 07:45:30 audit[1148]: pid=1148 uid=user2 logout",,,,,,
"Sep 15 07:47:30 audit[1946]: pid=1946 uid=guest selinux enforcement enabled",,,,,,
"Sep 15 07:49:30 audit[1450]: pid=1450 uid=admin login success",,,,,,
"Sep 15 07:51:30 audit[7841]: pid=7841 uid=user2 login success",,,,,,
"Sep 15 07:53:30 audit[5757]: pid=5757 uid=user1 login success",,,,,,
"Sep 15 07:55:30 audit[8502]: pid=8502 uid=user1 sudo CMD=/bin/ls",,,,,,
"Sep 15 07:57:30 audit[5202]: pid=5202 uid=user1 sudo CMD=/bin/ls",,,,,,
"Sep 15 07:59:30 audit[8987]: pid=8987 uid=admin accessed /etc/shadow",,,,,,https://we
.tl/t-NFpsU24grV
"Sep 15 08:01:30 audit[1916]: pid=1916 uid=user1 sudo CMD=systemctl restart ssh",,,,,,
"Sep 15 08:03:30 audit[3615]: pid=3615 uid=admin sudo CMD=systemctl restart ssh",,,,,,
"Sep 15 08:05:30 audit[3592]: pid=3592 uid=guest login success",,,,,,
"Sep 15 08:07:30 audit[7972]: pid=7972 uid=user2 logout",,,,,,
```

- While analyzing the CSV logs they discovered another link, which they pasted into the browser; the linked resource contained a text file encoded with a key.

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.



- Here, they found file name is hide 1 stegno pass .txt after they download it.



- They found encode message after they decode this message but they found something missing because they can't find properly.

Digital Trapdoor: Multi-layer Security using Encryption & Steganography.

Last build: 3 months ago - version 1.0 is here! Read about the new features [here](#) Options

463 Recipe ^ [Save] [Folder] [Trash] Input

To Base62 ^ [Pause] [Stop]

Alphabet
0-9A-Za-z

To Base64 ^ [Pause] [Stop]

Alphabet
A-Za-z0-9+/=

From Base64 ^ [Pause] [Stop]

Alphabet
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

rec 13 1

Output

lSpwdaBePHEni

2u1rVIMrvr28gesxj7

- They tried to decode the key but they can't find and try to find second missing part.

```
user@linux:~$ sudo iptables -L -n --line-numbers
user@linux:~$ ping 192.168.1.126
INPUT (policy ACCEPT)
           source      destination
1  DROP      192.168.1.16      0.0.0.0/0
FORWARD (policy ACCEPT)

OUTPUT (policy ACCEPT)

ping:192.168.1.126
ping: 192.168.1.126: No route to host
user@linux:~
```

- They discover another file at that time; system will be their IP is blocked or they might be dropped.
- They try to ping the system ip but they found error no route to host.

```
employee01@ubuntu:~$ sudo ufw status
verbose
Status: active
```

To	Action	From
--	-----	----
22	ALLOW	Anywhere
Anywhere	DENY	124.123 456.789

- When client check the ufw status attacker Ip will deny means Ip was blocked by the system.

7. Future Scope

Digital Trapdoor” project effectively combines encryption and steganography to secure data transmission; there remains significant potential for improvement and adaptation to modern cybersecurity challenges. The following enhancements outline possible directions for future development:

1. Integration of Advanced Cryptographic Algorithms

Future implementations can incorporate Elliptic Curve Cryptography (ECC), Homomorphic Encryption, or Quantum-Resistant Algorithms (like lattice-based or hash-base cryptography).

2. Use of Artificial Intelligence (AI) and Machine Learning (ML)

AI and ML can revolutionize the system by:

- Optimizing embedding strategies — using neural networks to identify optimal pixels or frequency regions for steganographic embedding without degrading carrier quality.

3. Cloud-Based Secure Communication

Integrating with cloud infrastructure (e.g., AWS, Azure, Google Cloud) will allow:

- Secure uploading and sharing of encrypted and embedded files via cloud storage.

4. Real-Time Secure Messaging and File Sharing

The system can be extended into a real-time encrypted communication platform featuring:

- Secure text, voice, and file transfer with steganography for hidden metadata or message tracking.
- Peer-to-peer (P2P) architecture with end-to-end encryption.
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5. Multi-Media Steganography

- Audio steganography using phase coding, echo hiding, or LSB coding in sound files.
- Video steganography by embedding data within motion vectors or discrete cosine transform (DCT) coefficients.

8. References

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